

OXYPC INVENTORY LIFECYCLE

BUSINESS

Business Requirements (BRD)

What the business needs from the app: objectives, scope, numbered requirements, risks and sign-off, with the business rules and SOPs each team follows.

Last updated: 6 Nov 2026 · Source: staging branch @ 55ce58a

App Development Team · Bluemonk

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In this document: Part A — Business Requirements · Part B — Business Rules & SOPs

PART A

Business Requirements

This document states, in business language, what the OxyPC Inventory Lifecycle app must do for Bluemonk's refurbished-device business: from buying a lot of used devices, through inspection, repair, testing and cosmetic finishing, to sale, dispatch and returns. It is the umbrella over the product and architecture documents in this folder and records which requirements are built, partly built, planned, or still a gap. It is written for the Director, TRC Manager, Sales Head, Accounts and team leads, not for developers.

1. Document purpose and audience

The BRD gives the business one agreed list of what the app must do, so that every change, UAT cycle and sign-off can be traced to a business need. It replaces `OxyPC_BRD_v6_Complete.docx` and is derived from the code on the `staging` branch, not from intentions. Where the business expects something the app does not do, the register says so.

Reader	Why they read it
Director	Approve scope, priorities and open decisions; sign off releases
TRC Manager	Confirm the floor process (inventory, parts, production, audits) is right
Sales Head	Confirm sales, dispatch and returns rules and the sale-from-any-stage decision
Accounts Head	Confirm invoice, payment, credit-note and e-way bill requirements
Production Manager and team leads	Check their daily work is covered and prioritised
Application Admin	Use the register as the source for PRDs, UAT scenarios and the roadmap

The **Product Vision** says why and the **Roadmap** says when; this BRD says what, with a status per requirement. The **Business Rules and SOPs** list the rules the code enforces today (rule IDs BR-01 to BR-114); this BRD groups them into requirements (IDs BR-001 onward, a separate numbering) and adds what is missing. The **PRDs** are the detailed specifications behind the "Planned" rows. The **Permission Matrix**, **KPI Definitions**, **State Machine**, **Security**, **Integration** and **Non-Functional** documents are the evidence behind each status, and the **UAT Plan** is how each requirement is accepted.

2. Business background and problem statement

2.1 What the business does

OxyPC Computers, run by Bluemonk, buys used laptops, desktops and TFT monitors in lots, refurbishes them and sells them. Each device receives a unique Tag Number at intake and is tracked by that tag for

life. One device can pass through up to ten teams:

1. **Lot intake and GRN.** A lot is bought, received against a Goods Receipt Note and registered device by device.
2. **IQC.** Each device is inspected; specification, condition and parts required are recorded. Mapping the GRN moves the tag to Stock In.
3. **Stock In and allocation.** The Inventory Manager holds stock and locations, then moves tags to the Production Manager, who allocates work.
4. **Repair L1–L4.** L1/L2 handles routine repair; hard cases go to L3/L4 and come back. Engineers raise part requests and mark Part Not Available (PNA).
5. **Stress Test.** A 27-point checklist, OS install and diagnostics. Fail sends the tag back to L1/L2.
6. **Cosmetic.** Cleaning, Putty, Dry Sanding, Masking, Painting, Water Sanding.
7. **Final QC.** Manual diagnosis and grading. Pass releases to Ready to Sale; fail routes back by reason.
8. **Sale and gate pass.** Pricing, block tags, sale, tax invoice and delivery challan, gate exit.
9. **Returns.** Internal return, paid repair, replacement or credit note; the tag re-enters stock.

Beside the device flow runs the **parts store**: new parts are booked through a Parts GRN, parts harvested from scrapped devices are booked as Harvest stock, and engineers' requests are handed over, verified, sourced or declared not in stock.

2.2 Why the app exists

Before the app, the story of a device lived on paper job cards, WhatsApp messages and spreadsheets. Nobody could answer "where is this tag, why is it there and what is it waiting for" without phoning a team lead. The app makes the tag the single unit of truth: every stage move, assignment (WorkID), part request, test result, grade, price, sale and return is recorded against it, and dashboards and reports are built from the same records.

2.3 What still hurts

- Engineers cannot see where a part request stands; PNA is tracked apart from the request.
- Fail reasons do not travel with the tag; each fail overwrites the last one.
- The cosmetic chain moves one tag at a time.
- The Final QC grade does not influence the selling price.
- Harvested parts are not offered before new parts.
- Invoice and e-way bill are not produced from the sale; Accounts re-keys sales into a billing tool.
- The TRC Manager has no single view of inventory, parts and production.
- Security and data-safety gaps: uploads served without login, default secrets, permissive role defaults, backups on the same server, uploads not backed up.

3. Business objectives and success metrics

Targets marked "proposed" need sign-off by the KPI owner named in the KPI Definitions; baselines are measured first (Roadmap N6).

Objective	KPI	Target / measure
O1 Faster flow	Median days from Stock In to Ready to Sale	−20% vs baseline (proposed, Q2 2027)

Objective	KPI	Target / measure
O1 Faster flow	Tags older than 30 days in one stage	−50% vs baseline (proposed, Q2 2027)
O2 Parts never block silently	Median PNA wait time (mark to clear)	≤ 5 working days (proposed, Q1 2027)
O2 Parts never block silently	Parts fill rate: handed over within 1 working day	≥ 85% (proposed, Q1 2027)
O2 Use what we own	Share of handed-over parts from Harvest stock	+25% vs baseline (proposed, Q2 2027)
O3 Right first time	First-pass yield at Stress Test and Final QC	≥ 85% at Final QC (proposed, Q2 2027)
O3 Right first time	Fails whose reason is still visible at the next stage	100% (today 0% for Stress notes)
O4 Price by grade	Ready to Sale tags with a grade-based suggested price	100% of graded tags (Q1 2027)
O5 Clean billing	Sales with an invoice number; sales above threshold with an e-way bill number	100% (Q1 2027); 100% (Q2 2027, no field today)
O6 Trust the numbers	Tags at Sold with no sale record ("orphans")	0 (Q4 2026)
O6 Trust the numbers	Writes covered by the audit log	≥ 90% (today about 35%)
O6 Trust the numbers	Daily Stock reconciliation	Closing = Opening + In − Out every day
O7 Fast pages	Dashboard loads over 1 second	< 5% of loads (Q4 2026)
O8 Adoption	Personas with signed UAT for their journey	12 of 12 per release
O9 Profitability	Lot P&L margin	≥ 15% (price-matrix default; to be agreed)

4. Scope

4.1 In scope

Area	Modules and processes
Intake	Lots, GRN with invoice, GRN post-IQC, IQC line items, OxyQC agent, bulk CSV upload, entity movement (ReGRN)
Inventory	Stock In, buckets, urgent flags, transfers, locations, physical and zone audits, trash and restore
Production and repair	Allocation, WorkIDs, L1/L2 and L3/L4 repair, part estimation, change engineer, scrap decisions
Parts store	Part requests, handover and verify, Not In Stock, Procure, PNA, Parts GRN (As New), Harvest, warranty parts
Quality	Stress Test checklist, cosmetic chain and flow data, Final QC pass/fail, grading, fail routing
Sales and dispatch	Ready to Sale pricing, block tags, sale, tax invoice and challan print, transport, gate pass, dispatch approval
Returns and warranty	Internal return, external tag (paid repair), replacement, credit note, return stock, extended warranty
Accounts	Customer receipts, supplier payments, invoice upload, sales list, daily stock

Area	Modules and processes
Reporting	Dashboard, Business P&L, Lot P&L, Daily Stock, Stock Aging, Overdue, Receivables, Sales Report, Stage Movement, WorkID Status
Administration	Users, roles, permission matrix, master data, cost config, stage control, tickets, backups
Planned	PRD-01 to PRD-04, batch cosmetic moves, harvest-first, one TRC view

4.2 Out of scope

Area	Reason
E-way bill generation (outbound)	Not built; only inbound numbers and files are stored. A future item under PRD-04 phase 2, not a requirement of the current system
GST e-invoice (IRN)	Not built; needs an Accounts decision and a GST provider account
Rentals	No rental feature exists; needs discovery first (Roadmap L5)
Trade Partner portal and external bidding	Per decision, outside the refurbishment flow covered here
Customer Care (tickets, pairing, offers)	Per decision, a pilot outside this BRD; its WhatsApp dependency was removed
CRM sourcing, telecalling, dealer calling; Reliance FieldOps	Separate product lines with their own UAT; FieldOps touches no OxyPC tables
Full accounting ledger; multi-tenant SaaS; WhatsApp and email	Accounts uses an external tool; not in this horizon; WhatsApp removed, no email service found

5. Stakeholders and roles

Four of the twelve personas (Production Manager, Final QC Inspector, Accounts, TRC Manager) have no built-in role today and run as custom roles through the permission matrix.

Role	Responsibility in the process	Interest in this BRD
IQC Inspector	Creates lots, inspects incoming devices, assigns tags, maps tags to the GRN	Intake rules; whether IQC may run GRN
Inventory Manager	Stock In, floor movement, locations, block tags, scrap, Ready to Sale release, warranties	Inventory, return stock and audit rules
Production Manager	Plans lots, allocates tags to engineers, reassigns work, tracks PNA and productivity	Production and PNA rules; a real role in the app
Repair Engineer (L1/L2, L3/L4)	Diagnoses and repairs, raises part requests, marks PNA, escalates, completes or scraps	Part status and fail history
QC Inspector (Stress Test)	Runs checklist, OS install and diagnostics; completes or fails	Stress Test and fail-reason rules
Cosmetic Lead	Runs the six-step chain; completes or fails each tag	Batch moves and worker credit
Final QC Inspector	Manual diagnosis, grade, pass to Ready or fail back by reason	Grade scale, routing, grade pricing
Spare Parts Manager	Books new and harvested parts, fulfils requests, sourcing and warranty parts	Parts store, PNA, harvest-first
Sales Executive	Sells Ready stock, blocks tags, records sales, prints challans, gate exit, returns	Sales and returns rules; sale-from-any-stage decision

Role	Responsibility in the process	Interest in this BRD
Accounts	Payments, invoices, sales list, credit notes, daily stock	Billing and e-way bill rules; an Accounts role
Managers / Admin	Dashboard, reports, P&L, tickets, users, roles, master data	KPIs, administration, security, audit
TRC Manager	Oversees inventory, parts and production; warranty extensions; daily stock and audits	One TRC view, audits, warranty

6. Business process overview

A tag enters at **GRN Receipt** (registered on a lot) or directly at **IQC**. Mapping or validating the GRN moves it to **Stock In**; the Inventory Manager moves it to **TRC Production**, where the Production Manager assigns it to L1/L2 Repair, Stress Test or Cosmetic.

In **L1/L2** the engineer picks the tag (opening a WorkID), repairs it, raises part requests and may escalate to **L3/L4**, which always hands the tag back to L1/L2, completed or with a scrap decision. Mark Complete is refused while any PNA is open; on completion the tag goes to Stress Test. Three quality gates follow, each able to complete forward or fail back:

Gate	Completes forward to	Fails back to
Stress Test	Cosmetic Received	L1/L2, with an engineer assigned and failed items saved as notes
Cosmetic chain (Received, Cleaning, Putty, Dry Sanding, Masking, Painting, Water Sanding, Completed)	Next step, or Skip straight to Final QC	L1/L2 (allowed at Received, Cleaning, Water Sanding, Completed)
Final QC	Pass Hold, then Move to Inventory releases the graded tag to Ready to Sale	Fail Hold, then by reason: Hardware to L1/L2 (TRC Production if no engineer), Software to Stress Test, Cosmetic to Cosmetic Received

From **Ready to Sale** the tag is priced, optionally blocked for a customer, and sold. The sale freezes the selling company details, draws a sale number from one sequence and computes warranty expiry; the tax invoice and delivery challan are printed and the gate confirms exit. By Director decision of 19 Oct 2026 a sale is allowed from any stage; only blocked tags are refused.

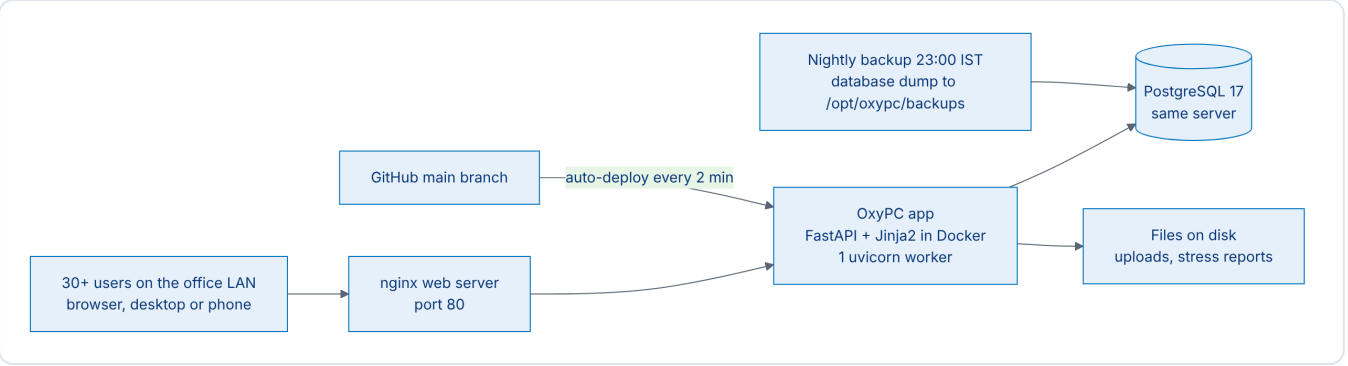
Parts inward: new parts arrive through a **Parts GRN (As New)** with PO, invoice, e-way bill and vehicle details; parts from scrapped devices are booked through **Harvest**. Requests are handed over, verified on the device, declared Not In Stock, or sent to Procure (a sourcing request closed in the CRM and verified into stock).

Returns: a return is logged against the sale with reason, condition and receiver; the tag moves to **Stock In** at once and waits for manager approval, which sends it to IQC (restock) or Scrapped. A replacement sells a Ready tag to the same customer at zero; a credit note needs a debit note number and advances through six stages. The app's "Returned" stage is never used.

7. Current solution setup and technologies

This section records how the app is set up today and what it is built with, so that business readers can see the shape of the solution without reading the Technical Architecture document. Detail is in Technical Architecture (Part A) and Operations & Testing (Part A).

7.1 Setup at a glance



Item	Current state
Where it runs	One Linux server on the office network (Internal Server), Asia/Kolkata time; not reachable from the internet
How users reach it	Browser over the LAN through nginx on port 80; the app itself listens on 127.0.0.1:8000 inside Docker
Application	Single Python web application (monolith): FastAPI serves both the pages (server-rendered Jinja2 templates) and the JSON endpoints used by the pages
Database	PostgreSQL 17 installed natively on the same server; schema changes applied with Alembic migrations (36 revisions) plus standalone migrate scripts
Files	Uploaded documents, photos and stress-test reports are stored on the server disk (bind-mounted folders), not in the database
Background work	Scheduled jobs (ageing, reminders, exports) run inside the app process; there is no separate job server or queue
Backup	Nightly database dump at 23:00 IST to a folder on the same server; uploads are not included; restore has not been rehearsed
Release path	Code is pushed to the staging branch on GitHub, reviewed, merged to main; a timer on the server pulls main every 2 minutes, rebuilds the Docker image, restarts the container and checks /health
Environments	Production (Internal Server) and developer machines only; there is no separate staging server (UAT is done on the production server or a developer laptop)
Legacy	A Windows installer build (PyInstaller + Inno Setup) and a bare-metal Linux service exist for older deployments and as a rollback path
Other systems	Partner portal app (separate container, port 3001), WhatsApp service (`wa-service`), field-ops app and QC agent are separate components integrated through the same database or HTTP calls

7.2 Technologies used

Layer	Technology	Version	Notes
Language	Python	3.12 (Docker image)	Windows scripts and bundle use 3.11–3.14
Web framework	FastAPI	0.115.6	Routers grouped by business domain
Web server	uvicorn	0.32.1	One worker behind nginx
Page templates	Jinja2	3.1.4	Server-rendered HTML pages
Database	PostgreSQL	17	16.6 in the Windows bundle

Layer	Technology	Version	Notes
Database access	SQLAlchemy (async) + asyncpg	2.0.36 / 0.30.0	ORM over 147 tables
Schema migrations	Alembic	1.14.0	36 revisions
Login and passwords	python-jose (JWT), bcrypt	3.3.0 / 5.0.0	Cookie-based sessions
Rate limiting	slowapi	0.1.9	Login and sensitive endpoints
Configuration	python-dotenv, `.env` file	1.0.1	Secrets must be set per server (see risks)
Excel import/export	openpyxl	≥3.1.0	Bulk uploads, reports
PDF and labels	fpdf2, reportlab, pypdf, pdfplumber, python-barcode	floor / unpinned	Gate pass, invoices, tag labels
PowerPoint export	python-pptx	≥1.0.0	Reports
Front end	Bootstrap, Bootstrap Icons	5.3.2 / 1.11.3	Served from `static/vendor` (no CDN)
Tables and widgets	jQuery, DataTables, Tom Select, Chart.js	3.7.1 / 1.13.7 / 2.3.1 / 4.4.0	Lists, filters, dashboard charts
Container	Docker Engine, Docker Compose	29.8.0 / 5.5.1	`python:3.12-slim` image, non-root user
Reverse proxy	nginx	not recorded	Serves `/static/` with 30-day cache
Source control and CI	GitHub, GitHub Actions	—	Tests run on push; deploy by server-side timer
Testing	pytest, pytest-asyncio, httpx	8.3.4 / 0.24.0 / 0.28.1	Plus UAT Excel runner
Legacy packaging	PyInstaller, Inno Setup 6	≥6.10.0	Windows installer build

8. Business requirements register

Priority: **Must** (the business cannot run without it), **Should** (agreed opportunity), **Could** (desirable). Status: **Built**, **Partial** (exists with known gaps), **Planned** (in a PRD or roadmap item), **Gap** (expected, nothing in the code, not yet planned).

7.1 Intake and IQC

ID	Area	Requirement	Priority	Status	Source
BR-001	Intake & IQC	The system shall record every lot with supplier, invoice, quantity, buying price and inbound e-way bill.	Must	Built	Rules BR-09; UAT E2E-01.1
BR-002	Intake & IQC	The system shall give every device a unique Tag Number and refuse duplicates at IQC.	Must	Built	Rules BR-01
BR-003	Intake & IQC	The system shall record specification, condition, grade, location and parts required at IQC, auto-filled from the OxyQC agent, with bulk CSV upload and a per-row error list.	Must	Built	Rules BR-03; Integration 4.1, 7.1

ID	Area	Requirement	Priority	Status	Source
BR-004	Intake & IQC	The system shall move IQC tags to Stock In when their GRN is mapped or validated and warn on quantity mismatch.	Must	Built	Rules BR-06, BR-08, BR-09
BR-005	Intake & IQC	The system shall let the IQC Inspector run the GRN screens for the lots they inspect.	Should	Gap	Rules OQ4; State Machine Q8

7.2 Inventory and Stock

ID	Area	Requirement	Priority	Status	Source
BR-006	Inventory & Stock	The system shall allow only Stock In tags to move to TRC Production or L1/L2, singly or by bucket.	Must	Built	Rules BR-12, BR-14
BR-007	Inventory & Stock	The system shall log every floor transfer with employee and location without changing the stage, and require Authorized By, dates and reason on an urgent flag.	Must	Built	Rules BR-16 to BR-22
BR-008	Inventory & Stock	The system shall support physical and zone audits recording system versus seen stage and location, once per tag per date.	Must	Built	Rules BR-110
BR-009	Inventory & Stock	The system shall write a stage-movement record for every stage change.	Must	Partial	Rules BR-111; State Machine Q12 (two silent moves)
BR-010	Inventory & Stock	The system shall allow a sold tag to return to Ready to Sale under a new GRN, invoice and entity (ReGRN).	Should	Built	Rules BR-25

7.3 Production and Repair

ID	Area	Requirement	Priority	Status	Source
BR-011	Production & Repair	The system shall let the Production Manager assign tags or buckets to L1/L2, Stress Test or Cosmetic and to a named engineer, opening a WorkID and closing the old one on every reassignment or pick.	Must	Built	Rules BR-26, BR-27, BR-31, BR-113
BR-012	Production & Repair	The system shall show engineers their own tags plus the unclaimed pool, and supervisors the full queue.	Must	Built	Rules BR-32
BR-013	Production & Repair	The system shall support escalation to L3/L4 and return the tag to L1/L2 on completion or scrap decision (Normal or Replacement).	Must	Built	Rules BR-33 to BR-36
BR-014	Production & Repair	The system shall refuse L1/L2 Mark Complete while a PNA is open, then auto-verify handed-over parts and cancel the rest.	Must	Built	Rules BR-37, BR-38
BR-015	Production & Repair	The system shall estimate a lot's parts cost from IQC-required parts and raise a lot sourcing request from it.	Should	Built	Rules BR-43
BR-016	Production & Repair	The system shall warn at 70% and auto-scrap at 100% of expected sale value.	Could	Partial	Rules BR-40; PRD-03 (expected value never set)
BR-017	Production & Repair	The system shall check the tag's current stage and the user's role before every stage move.	Must	Gap	Key Findings P2, P3; State Machine Q5
BR-018	Production & Repair	The system shall report output per engineer and team from WorkIDs, with ageing and CSV export.	Should	Built	KPI O1, O5

7.4 Parts Store

ID	Area	Requirement	Priority	Status	Source
BR-019	Parts Store	The system shall let engineers raise part requests (New, Replace, Downgrade) linked to the open WorkID, one row per part, singly or in bulk.	Must	Built	Rules BR-67, BR-68
BR-020	Parts Store	The system shall support Handover, Not In Stock, Procure, Scrap and warranty send/receive, with handover quantity equal to requested.	Must	Built	Rules BR-69 to BR-74
BR-021	Parts Store	The system shall book new parts through a Parts GRN with PO, invoice, e-way bill and vehicle details into Part Master stock.	Must	Built	Rules BR-76
BR-022	Parts Store	The system shall book harvested parts as Harvest stock linked to the source tag.	Must	Partial	Rules BR-77, OQ9 (no tag link)
BR-023	Parts Store	The system shall show each request's status on the engineer's queue, notify the engineer on handover, link PNA to its request and keep every mark/clear cycle.	Should	Planned	PRD-01; Roadmap N2
BR-024	Parts Store	The system shall suggest Harvest stock before New stock at handover and report the harvest share.	Should	Planned	Roadmap X3
BR-025	Parts Store	The system shall show low-stock alerts, parts used today, stock value and New versus Harvest counts.	Should	Built	KPI Q8, O3

7.5 Quality (Stress, Cosmetic, Final QC)

ID	Area	Requirement	Priority	Status	Source
BR-026	Quality	The system shall provide the 27-item Stress checklist, save one report per visit, and move the tag to Cosmetic on Complete or to an assigned L1/L2 engineer on Fail.	Must	Built	Rules BR-44 to BR-47
BR-027	Quality	The system shall run the cosmetic chain in fixed order with an assignee on every move, allow Skip to Final QC, and allow Cosmetic Fail to L1/L2 at Received, Cleaning, Water Sanding and Completed.	Must	Built	Rules BR-49 to BR-53
BR-028	Quality	The system shall let the Cosmetic Lead advance many tags one step in one action with a per-tag result.	Should	Planned	Roadmap X2
BR-029	Quality	The system shall require Final QC Pass to set the grade, hold the tag until Move to Inventory, and finalise device cost once (buying price plus parts).	Must	Built	Rules BR-58, BR-59, BR-64
BR-030	Quality	The system shall require a reason on Final QC Fail and route by reason to L1/L2, Stress Test or Cosmetic Received, to the last person on that step.	Must	Built	Rules BR-61 to BR-63
BR-031	Quality	The system shall keep every fail (stage, reason, items, who, when, routed to) as history on the tag and show the open fail on the receiving queue, never overwriting an earlier fail.	Must	Planned	PRD-02 phase 1; Roadmap N3
BR-032	Quality	The system shall use one reason list and one routing rule for Stress, Cosmetic and Final QC fails and show a rework count per tag.	Should	Planned	PRD-02 phase 2; Roadmap X6
BR-033	Quality	The system shall use one agreed grade scale across Final QC, the device record, Sales and CRM.	Must	Gap	Key Findings P4; Vision Q5

7.6 Sales and Dispatch

ID	Area	Requirement	Priority	Status	Source
BR-034	Sales & Dispatch	The system shall compute a minimum Ready to Sale price from cost, other rate and margin, and allow manual prices only with the Set Price permission.	Must	Built	Rules BR-88; PRD-03
BR-035	Sales & Dispatch	The system shall suggest a min/max price from grade and device type, keep the cost floor as a warning, and audit Sales overrides with a reason.	Should	Planned	PRD-03; Roadmap X1
BR-036	Sales & Dispatch	The system shall let staff block Ready tags for a customer or colleague and refuse a sale of a blocked tag except from its own block.	Must	Built	Rules BR-79, BR-87
BR-037	Sales & Dispatch	The system shall check the Ready to Sale stage on the server when a block is created and audit block and release.	Should	Gap	Rules OQ11; State Machine Q11
BR-038	Sales & Dispatch	The system shall sell only tags of the seller's entity, number sales from one sequence, freeze company details on the sale, compute warranty expiry, and let only Admin delete or re-date a sale.	Must	Built	Rules BR-78, BR-82 to BR-86
BR-039	Sales & Dispatch	The system shall allow a sale from any stage (Director decision, 19 Oct 2026), refusing only blocked tags, and warn below cost.	Must	Built	Rules BR-80, BR-81; Key Findings P1 (to be confirmed, section 11)
BR-040	Sales & Dispatch	The system shall print a GST tax invoice and delivery challan from the frozen company snapshot with intra/inter-state tax split.	Must	Partial	Integration 2.1, 2.2; Key Findings F2 (one tag per invoice, HSN fixed, free-text number)
BR-041	Sales & Dispatch	The system shall produce one invoice per sale event covering all tags, with a server-generated number series per company and financial year.	Should	Planned	PRD-04 phase 1; Roadmap X4
BR-042	Sales & Dispatch	The system shall capture the outbound e-way bill number, date, vehicle and transporter on the sale, print them on challan and gate pass, and warn or block gate exit when they are missing.	Should	Planned	PRD-04 US-03, US-04
BR-043	Sales & Dispatch	The system shall generate the outbound e-way bill (and e-invoice if required) through a GST provider.	Could	Gap	Key Findings F1; Integration 2.3; out of scope (section 4.2)
BR-044	Sales & Dispatch	The system shall let the gate confirm exit per sale, singly or in bulk, marking the sale dispatched.	Must	Built	Rules BR-89; Integration 2.5
BR-045	Sales & Dispatch	The system shall require Sales Manager approval of a telecaller's dispatch request before goods leave.	Should	Partial	Rules BR-90, OQ13 (sale does not check it)

7.7 Returns and Warranty

ID	Area	Requirement	Priority	Status	Source
BR-046	Returns & Warranty	The system shall log a return against a sale with reason, condition and receiver, allow one return per sale, compute warranty status on the server and move the tag to Stock In pending approval.	Must	Built	Rules BR-91 to BR-93, BR-96

ID	Area	Requirement	Priority	Status	Source
BR-047	Returns & Warranty	The system shall support replacement at zero price, external tag return with paid-repair amount, credit notes through six stages, and Return Stock re-sale to the original customer.	Must	Built	Rules BR-94, BR-97 to BR-100
BR-048	Returns & Warranty	The system shall let Admin or Sales Manager approve a return to IQC or Scrapped, and define what happens to a rejected return.	Must	Partial	Rules BR-95, OQ2 (rejected tag stays in Stock In)
BR-049	Returns & Warranty	The system shall set or extend a warranty only on a sold tag, later than the current expiry, keeping a running total of added days.	Should	Built	Rules BR-105, BR-106

7.8 Accounts

ID	Area	Requirement	Priority	Status	Source
BR-050	Accounts	The system shall record customer receipts against sales and supplier payments against lots or POs, with invoice upload, and show paid and due per sale with receivables ageing.	Must	Built	Rules BR-101; KPI Q7, R12
BR-051	Accounts	The system shall give a dedicated Accounts role access to the Accounts pages through the permission matrix, not a fixed role list.	Must	Gap	Key Findings (roles not in code); Rules OQ3
BR-052	Accounts	The system shall produce the Daily Stock report (opening, closing, net per stage, sold today) on screen and as files every evening.	Must	Built	Rules BR-104; KPI R8

7.9 Reporting and Dashboard

ID	Area	Requirement	Priority	Status	Source
BR-053	Reporting & Dashboard	The system shall show a dashboard (stage pipeline, financial-year summary, category tiles, charts, role queues) and Business and Lot P&L, where every tile matches its page and the same calculation is used throughout.	Must	Partial	KPI K1 to K6 (revenue ignores overrides; active lots counts trashed; two P&L calculations)
BR-054	Reporting & Dashboard	The system shall provide Stock Aging, Overdue, Sales Report with orphan check and Stage Movement log, with export.	Must	Built	KPI R10 to R14
BR-055	Reporting & Dashboard	The system shall provide a Flow KPIs page (stage ageing, first-pass yield, PNA wait, fill rate, grade mix, sales cycle time) with drill-down.	Should	Planned	KPI section 5; Roadmap X5
BR-056	Reporting & Dashboard	The system shall provide one TRC view combining tags by stage and age, PNA, open requests, Final QC fails, daily stock and last audit.	Should	Planned	Roadmap L1

7.10 Administration and Access

ID	Area	Requirement	Priority	Status	Source
BR-057	Administration & Access	The system shall let Admin maintain users, roles, master data, cost configuration, tickets and the permission matrix that drives sidebar and page access.	Must	Built	Permission Matrix section 1

ID	Area	Requirement	Priority	Status	Source
BR-058	Administration & Access	The system shall provide a role for each of the twelve personas, including Production Manager, Final QC Inspector, Accounts and TRC Manager.	Must	Gap	Key Findings (roles not in code)
BR-059	Administration & Access	The system shall gate every page and button by the permission matrix, never by hard-coded role names.	Must	Partial	Security R9; NFR-09 (15 templates)
BR-060	Administration & Access	The system shall hide prices from users without Pricing Visibility on screens and in all exports.	Must	Partial	Key Findings S5; Security R6; see BR-067 to BR-070
BR-061	Administration & Access	The system shall keep one consistent list of stages and allowed moves, editable only by Admin.	Must	Partial	Rules BR-112; Key Findings P3
BR-067	Administration & Access	The system shall show buying price, cost, margin and P&L only to management roles (Admin, Sub Admin, TRC Manager, Managers), everywhere: screens, JSON, exports, printouts and reports.	Must	Gap	Director policy 1 Nov 2026; Security R3/R6/R9
BR-068	Administration & Access	The system shall default Pricing Visibility to off for every new or custom role and require an Admin to grant it.	Must	Gap	Director policy 1 Nov 2026; Security R13
BR-069	Administration & Access	The system shall apply pricing visibility to all CSV/Excel/PDF exports and printouts.	Must	Gap	Director policy 1 Nov 2026; Security R6
BR-070	Administration & Access	The system shall allow Sales Manager and Accounts to see selling price (and Sales Manager to set it) without seeing buying price, cost or margin.	Should	Partial	Director policy 1 Nov 2026; Rules BR-88; Permission Matrix section 4
BR-071	Administration & Access	The system shall let the Production Manager see parts cost for lot estimation and repair budgets without seeing buying price, selling price, margin or P&L.	Should	Gap	Director policy 1 Nov 2026; Security section 9.1

7.11 Audit and Data Safety

ID	Area	Requirement	Priority	Status	Source
BR-062	Audit & Data Safety	The system shall audit every write that matters (stage moves, prices, sales, returns, parts, deletions) with user, time, old and new values.	Must	Partial	NFR-11; Vision G6 (about 35%)
BR-063	Audit & Data Safety	The system shall audit every permission change, role change and admin password reset.	Must	Gap	Key Findings S4; Security R5
BR-064	Audit & Data Safety	The system shall back up database and uploads daily with an off-server copy and a recorded monthly test restore.	Must	Partial	Key Findings D1, D2; NFR-06, NFR-07
BR-065	Audit & Data Safety	The system shall use soft delete with restore and exclude trashed records from every count and report.	Must	Partial	KPI K3; NFR-19
BR-066	Audit & Data Safety	The system shall keep the approved database diagram in step with the live schema.	Should	Gap	Key Findings housekeeping (58 of 147 tables missing)

9. Non-functional business requirements

ID	Area	Requirement	Priority	Status
NF-01	Availability	Available in business hours, unplanned downtime under 30 minutes per incident, self-restart after crash or reboot, monitored from outside the server.	Must	Partial (auto-restart; single server; no monitoring)
NF-02	Performance	List and dashboard pages respond within 2 seconds for 95% of requests at 30 users; large lists page on the server.	Should	Partial (13 lists paged; no baseline)
NF-03	Capacity	50 concurrent users without one user's long task (for example a backup) freezing others.	Should	Partial (one worker; backup runs inside the app)
NF-04	Backup	Database and uploads backed up daily, one copy off the server, daily check that it ran.	Must	Partial (database only; same server; no alert)
NF-05	Recovery	Agreed recovery targets (acceptable data loss and downtime) and a restore tested and recorded monthly.	Must	Gap (no targets; dump and restore script disagree)
NF-06	Access control	New custom roles denied by default until granted; sensitive modules (Cost Config, Credit Note, Permission Matrix, Users) never open by default.	Must	Gap (permissive default; auto-created sub_admin sees every non-admin page)
NF-07	Access control	Exports and prints respect Pricing Visibility and the Export, Download, Print and Upload switches; by Director policy of 1 Nov 2026, buying price, cost, margin and P&L are visible to management roles only, everywhere in the application.	Must	Partial (switches stored, not enforced; pricing visible to all roles by default)
NF-08	Security	Uploaded files (KYC, payment proofs, invoices, daily stock files) served only to logged-in users with permission.	Must	Gap (served at /uploads with no login)
NF-09	Security	Refuse to start in production with default secrets (database password, agent key, admin and UAT passwords); UAT accounts disabled in production.	Must	Gap (defaults ship in code and seed scripts)
NF-10	Security	HTTPS on the LAN; sessions expire after one day, lock after five failed logins, and end on password change.	Should	Partial (lockout and expiry in place; plain HTTP; no revocation)
NF-11	Audit trail	Audit before the change, append-only audit rows, agreed retention.	Must	Partial (about 35% coverage; no retention policy)
NF-12	Data integrity	Money totals computed on the server in exact decimals; schema changes approved on the diagram first.	Must	Partial (invoice GST computed in floating point)
NF-13	Release safety	No change reaches production without automated tests passing and staging UAT sign-off.	Must	Gap (no staging; tests not run automatically)
NF-14	Usability	Works on current Chrome and Edge at 90% zoom; gate and scanning pages usable on a phone.	Should	Partial (no phone layout for Gate Pass)

10. Assumptions, constraints and dependencies

10.1 Assumptions

- Where code and business view differ, the business view is the requirement and the difference is a Partial or Gap.
- The twelve personas are the complete list of business roles; one person may hold more than one.
- A number is trusted only when a tile and its page agree; KPI owners will sign the definitions.
- Accounts keeps the general ledger in an external tool; the app records payments and invoices only.
- Management roles for pricing/P&L are Admin, Sub Admin, TRC Manager and Managers; Sales Manager and Accounts see selling price only; Production Manager sees parts cost only, for lot estimation (confirmed by the Director, 1 Nov 2026).
- Current volumes (about 30 concurrent users; lots of 50 to 200 devices) hold for the next three quarters.

10.2 Constraints

- One physical server on the office LAN over plain HTTP holds app, database and backups.
- Schema changes need approval on the database diagram, so new tables are batched into one review per quarter.
- There is no staging server today; UAT cannot run safely until one exists (Roadmap N1).
- One development team maintains the app; roadmap estimates assume that capacity.
- GST, HSN and e-way bill rules must be confirmed by Accounts before any billing requirement is built; this document gives no tax advice.

10.3 Dependencies

- PRD-01 (part status) before harvest-first (BR-024) and the TRC view (BR-056).
- PRD-02 phase 1 (fail history) before rework routing (BR-032) and the Flow KPIs page (BR-055).
- PRD-03 (grade pricing) needs one agreed grade scale (BR-033) and an approved price table.
- PRD-04 phase 1 (one invoice per sale) before e-way bill capture; provider integration needs a GST provider account owned by Accounts.
- KPI targets in section 3 need measured baselines first (Roadmap N6).
- UAT sign-off per persona needs UAT accounts for the four custom roles on staging.

11. Risks and open decisions

Risk / decision	Impact	Owner	Mitigation / next step
Director decision of 19 Oct 2026: a tag may be sold from any stage, including Sold or in repair	Sales can bypass Final QC and grading; a sold tag can be sold twice; Ready to Sale counts lose meaning	Director, Sales Head	Confirm the decision; if kept, refuse a second sale of a Sold tag and audit every non-Ready sale; if not, restore the Ready to Sale check
Grade scale differs by screen	Grade pricing cannot be built; grades mean different things to different teams	Final QC lead, Sales Head	Agree one scale before PRD-03

Risk / decision	Impact	Owner	Mitigation / next step
Four personas have no role; Accounts pages refuse a custom Accounts role	Accounts staff work under a manager's login; access cannot be narrowed	Application Admin	Create the roles; replace fixed role lists with the matrix
Uploads served without login; default secrets; permissive role defaults	KYC, payment proofs and invoices readable on the network; unknown users could gain wide access	Application Admin	Serve uploads through a logged-in route; change defaults on the server; deny-by-default for new roles (fix first)
Pricing and P&L currently visible to all roles by default (policy: management only)	Buying cost and margin exposed to floor and sales staff	Director / Application Admin	Change defaults and gates per Security & Access Design (role list confirmed 1 Nov 2026)
Backups on the same server; uploads not backed up; no test restore; no recovery targets	A disk or server failure loses data and documents; a restore may fail when needed	Director, Application Admin	Agree recovery targets; off-site copy and uploads backup; run and record a test restore
No staging server; deploy goes straight to production; tests not run automatically	Untested changes reach floor staff	Application Admin	Stand up staging before the first UAT cycle (Roadmap N1); add an automated test run
Schema diagram stale (58 of 147 tables missing)	The schema approval rule cannot be followed	Application Admin	Regenerate from the live schema and approve once
Audit trail covers about 35% of writes; permission changes not audited	Disputes about who changed what cannot be settled	Managers / Admin	Each PRD adds audit actions; audit permission and password changes first
Dashboard figures disagree with reports	Managers lose trust in the numbers	Managers / Admin	Dashboard KPI hygiene (Roadmap N4); KPI owners sign definitions
E-invoice (IRN) needed as well as e-way bill?	Changes scope and provider for PRD-04 phase 2	Accounts Head	Decide before PRD-04 phase 2 is estimated
Rentals listed as a Sales task but no feature exists	Rental devices tracked outside the app	Sales Head	Discovery (Roadmap L5)
Rejected returns stay in Stock In; Returned stage unused	Rejected tags look restocked	Sales Head	Define the rule and update the returns flow

12. Acceptance and sign-off

A requirement is accepted when its UAT scenarios pass on staging as set out in the UAT Plan and Sign-off: the end-to-end journey (one tag from Lot to Credit Note) passes, every persona's scenarios run with at least 95% passing, no Critical or High defects remain open, dashboard and report figures reconcile for the test tags, and the release is marked Approved in the QA Dashboard. Planned requirements are accepted against the acceptance criteria in their PRD; Gap requirements only once a fix has been scheduled and tested the same way. Signing below confirms that the requirements and statuses in this document are a correct statement of the business need on the date given.

Role	Name	Signature	Date
Director			
TRC Manager			
Sales Head			
Accounts Head			

Role	Name	Signature	Date
Production Manager			
Application Admin			

13. Glossary

Term	Meaning
Tag / Tag Number	The unique barcode given to one device at intake; the unit of truth for its whole life
Lot	A batch of devices bought from one supplier under one invoice
GRN	Goods Receipt Note; the record that a lot or spare parts were received. Mapping a tag to its GRN moves it to Stock In
IQC	Incoming Quality Check; the first inspection recording specification, condition and parts required
Stock In	Inspected devices waiting to be allocated to production
TRC	The refurbishment centre (inventory, parts and production teams); TRC Production is its allocation stage
WorkID	One assignment of a tag to one person, opened on assignment and closed when the tag moves on
L1/L2	Routine repair level; every escalation or fail lands here
L3/L4	Advanced repair level; always hands the tag back to L1/L2
PNA	Part Not Available; a mark that a needed part cannot be supplied, which blocks Mark Complete
Part request	An engineer's request for a part, with statuses from Requested through Handed Over, Received, Not In Stock and Sent for Sourcing to Cancelled
Parts GRN (As New)	Record of new spare parts received from a vendor with PO, invoice and e-way bill
Harvest	A part recovered from an old or scrapped device and booked into Part Master
Stress Test	The 27-item QC checklist, OS install and diagnostics run after repair
Cosmetic chain	Cleaning, Putty, Dry Sanding, Masking, Painting, Water Sanding
Final QC	The last manual check that grades the device and passes it to Ready to Sale or fails it back by reason
Grade	Quality grade given at Final QC (A/B/C on screen; A–E plus Scrap in the device record)
Ready to Sale	Stock that passed Final QC and can be priced and sold ("Finish Good" in transfers)
Block Tags	Holding Ready tags for a staff member or customer so nobody else sells them
Entity	The selling company a tag belongs to; a seller sells only their own entity's tags
Gate pass	The security gate's confirmation, against the delivery challan, that sold goods have left
E-way bill	Government document for moving goods above a value threshold; captured inbound only today
Credit note	A return settled by money rather than replacement; six stages ending in Payment Invoice Done
Permission matrix	Admin screen saying which role can open which module; drives sidebar and page access

PART B

Business Rules & SOPs

1. Purpose

This document lists the business rules the OxyPC Inventory Lifecycle app enforces today, and a short daily routine (SOP) for each of the 12 business roles.

- Every rule cites the file and function it comes from. If a rule is not in the code, it is not in this list.
- "Role" is the business persona. Where the code checks a different role, the code role is shown in brackets.
- Where the business view and the code disagree, the gap is listed in section 4.

Stage names and transitions are in the Device Lifecycle State Machine.

1.1 Business roles and the code roles behind them

#	Business role	Built-in code role(s) today	Note
1	IQC Inspector	`iqc_inspector` ("IQC Handler")	Cannot open `/grn` unless the permission matrix grants it
2	Inventory Manager	`inventory_manager`	Also acts as Production Manager in code
3	Production Manager	`inventory_manager`, or a custom role `production_manager`	No built-in role (`routers/buckets.py::production_manager`)
4	Repair Engineer	`l1_engineer` ("L1/L2 Engineer"), `l2_engineer`, `l3_engineer` ("L3/L4 Engineer")	
5	QC Inspector	`qc_inspector` ("QC Handler")	
6	Cosmetic Lead	`cosmetic_manager`	Single-stage custom roles also exist
7	Final QC Inspector	Any role with `cosmetic_finalqc` edit permission	No built-in role
8	Spare Parts Manager	`spare_parts_manager` ("Parts Manager")	
9	Sales Executive	`sales` ("Sourcing Sales"), `telecaller`, `sales_manager`	
10	Accounts	None; Accounts pages allow `admin`, `inventory_manager`, `sales_manager`	`routers/accounts.py::FINANCE_ROLES`
11	Managers / Admin	`admin`	Admin passes every role check
12	TRC Manager	Custom role `trc_manager`	Sees the full repair queue (`routers/repair.py::FULL_QUEUE_ROLES`)

Custom (admin-created) roles pass any non-admin-only `require_roles` gate and are then limited by the Module Permission matrix (`auth/dependencies.py::require_roles`).

2. Business rules

2.1 Intake: Lot, GRN and IQC

ID	Rule	Source	Role
BR-01	A tag number (barcode) must be unique; IQC refuses a barcode that already exists.	`routers/iqc.py::iqc_create`	IQC Inspector
BR-02	An IQC entry with no lot is filed under the system "Unassigned" lot.	`routers/iqc.py::iqc_create`, `utils/lot_helpers.py::get_or_create_unassigned_lot`	IQC Inspector
BR-03	An IQC entry that carries a GRN number goes straight to Stock In; without one it stays at IQC.	`routers/iqc.py::iqc_create`	IQC Inspector
BR-04	The device price defaults to the lot line item's unit price, else the lot average (buying price ÷ qty).	`routers/iqc.py::iqc_create`	IQC Inspector
BR-05	Tags registered on a lot start at GRN Receipt and move to IQC with "Advance GRN to IQC".	`routers/stock.py::register_device_barcode`, `advance_grn_to_iqc`	IQC Inspector
BR-06	Mapping a GRN moves only tags at IQC to Stock In; tags in other stages just receive the GRN number.	`routers/grn.py::grn_map`	Inventory Manager (code: `admin`, `inventory_manager`)
BR-07	A tag with no entity gets "OxyPC Computers" when its GRN is mapped; an existing entity is never overwritten.	`routers/grn.py::grn_map`	Inventory Manager
BR-08	Validating a Plant GRN auto-maps every IQC-stage tag of the linked lot to Stock In.	`routers/grn.py::grn_validate`	Inventory Manager
BR-09	A lot GRN records expected vs received quantity and warns on a mismatch; the GRN number is auto-generated if blank.	`routers/grn.py::submit_grn`	Inventory Manager
BR-10	"Pending for TRC" bulk move sends tags back to IQC with the GRN cleared ("Sending for Selling") or to Stock In ("Sold to OxyPC"); tags whose move is not an allowed transition are skipped.	`routers/grn.py::grn_bulk_move_to_trc`	Inventory Manager
BR-11	"Yes As-Is" writes one Sub-Lot Number (and optional Assigned To) on every tag mapped to that GRN.	`routers/grn.py::grn_as_is_lot`	Inventory Manager

2.2 Stock and floor movement

ID	Rule	Source	Role
BR-12	Only tags at Stock In can be moved to TRC Production or to L1/L2.	`routers/stock.py::move_to_trc`, `_move_device_to_l1l2`	Inventory Manager
BR-13	A tag moved to L1/L2 from Stock In lands unassigned (Pick This), and any stale L1 WorkID is closed first.	`routers/stock.py::_move_device_to_l1l2`	Inventory Manager
BR-14	"Move to Production" on a bucket moves only its Stock In tags, except customer-return buckets, which move all their tags.	`routers/buckets.py::_move_bucket_devices_to_trc`	Inventory Manager
BR-15	A bucket name is reused case-insensitively; creating a bucket sets its tags to Stock In.	`routers/buckets.py::create_bucket`	Inventory Manager

ID	Rule	Source	Role
BR-16	An urgent flag needs all four fields: Authorized By, Request Date, Target Date and Reason.	`routers/stock.py::stock_flag_save`	Inventory Manager
BR-17	The urgent flag clears itself whenever a tag moves into Stock In.	`models/device.py::_clear_flag_on_stock_in`	System
BR-18	A transfer (Move Item / Bucket / Lot) records the assignment and location only; it never changes the stage.	`routers/transfers.py::create_transfer`, `_move_devices_bulk`	Inventory Manager
BR-19	Every transfer needs an employee to assign to and creates an `asgn` WorkID plus a notification.	`routers/transfers.py::create_transfer`	Inventory Manager
BR-20	A transfer of type Ready for Sale (shown as "Finish Good") or As-Is Lot stamps the employee as the tag's Assigned To.	`routers/transfers.py::create_transfer`	Inventory Manager
BR-21	A transfer of type Scrap for Sale creates one Scrap ID batch per submission.	`routers/transfers.py::_maybe_create_scrap_batch`	Inventory Manager
BR-22	A new Location ID resets the tag's Bin Location and writes a location log row.	`routers/transfers.py::create_transfer`	Inventory Manager
BR-23	The Customise modal can never set a tag to Sold; selling must go through New Sale.	`routers/iqc.py::iqc_bulk_apply_grade_type`	Inventory Manager
BR-24	Customise stage moves skip the allowed-transition check for admin and when used from All Inventory or Inventory Manager pages.	`routers/iqc.py::iqc_bulk_apply_grade_type`	Inventory Manager
BR-25	ReGRN (Entity Movement) moves a sold tag back to Ready to Sale under the new GRN, invoice and entity.	`routers/entity_movement.py` (change entities)	Inventory Manager / TRC Manager

2.3 Production and repair

ID	Rule	Source	Role
BR-26	Production assignment has three departments: L1/L2 Repair → L1/L2, Stress Test → Stress Test, Cosmetic Repair → Cosmetic Received.	`routers/buckets.py::DEPT_TO_STAGE`, `_apply_department_move`	Production Manager
BR-27	Reassigning a tag closes the previous person's open WorkID and opens a new one at the same moment.	`services/work_order_close.py::close_open_work_orders`	Production Manager
BR-28	Assigning a Stress Test engineer reuses the open, unassigned STRS- WorkID instead of creating a second one.	`services/work_order_close.py::reuse_open_stress_work_order`	Production Manager
BR-29	Change Engineer can move a tag from any stage to L1/L2, L3/L4, Stress Test, Cosmetic, Final QC, Stock In or TRC Production and assign any active user.	`routers/stock.py::change_engineer`, `CHANGE_ENGINEER_STAGES`	Production Manager
BR-30	Any tag entering L1/L2 from another stage starts a fresh repair cycle (status New, L3/L4 status cleared).	`models/device.py::reset_repair_cycle_on_entry`	System
BR-31	"Pick This" lets the clicking engineer take a tag from anyone; all open WorkIDs close and a new one opens.	`routers/repair.py::l1_pick`, `l3l4_pick`	Repair Engineer
BR-32	Engineers see their own tags plus the unclaimed pool; admin, TRC Manager, Inventory Manager and Production Manager see the full queue.	`routers/repair.py::FULL_QUEUE_ROLES`	Repair Engineer / supervisors

ID	Rule	Source	Role
BR-33	Request to L3/L4 moves the tag to L3/L4 and creates an L3L4- WorkID for the chosen engineer; the L1/L2 WorkID stays open.	`routers/repair.py::request_l3l4`	Repair Engineer (L1/L2)
BR-34	L3/L4 Complete and L3/L4 Scrap both send the tag back to L1/L2 for the next action.	`routers/repair.py::l3l4_complete`, `l3l4_scrap`	Repair Engineer (L3/L4)
BR-35	L3/L4 scrap must be "Normal Scrap" or "Replacement Scrap".	`routers/repair.py::l3l4_scrap`	Repair Engineer (L3/L4)
BR-36	After an L3/L4 scrap decision, "Back to Inventory" sets grade Scrap and moves the tag to Scrap for Sale (Tags Scrapped page).	`routers/repair.py::back_to_production`	Repair Engineer
BR-37	L1/L2 Mark Complete is refused while the tag has any active PNA part.	`routers/repair.py::l1l2_complete_to_stress`	Repair Engineer
BR-38	Mark Complete auto-verifies handed-over parts (cost counted) and auto-cancels parts never handed over.	`routers/repair.py::_mark_complete_open_parts`, `l1l2_complete_to_stress`	Repair Engineer
BR-39	Mark Complete always moves the tag to Stress Test and raises an STRS- WorkID; the Stress engineer is optional.	`routers/repair.py::l1l2_complete_to_stress`	Repair Engineer
BR-40	In the legacy Complete Job panel, a repair whose total cost reaches the expected sale value is auto-scrapped; at 70% the engineer is warned.	`services/cost_engine.py::check_scrap_decision`, `routers/repair.py::complete_repair`	System
BR-41	L2 repair needs an earlier L1 attempt and L3 needs an L2 attempt (legacy Start Repair).	`services/control_engine.py::validate_repair_level`	Repair Engineer
BR-42	The L2 stage is retired; every escalation or fail lands in L1/L2.	`routers/repair.py::complete_repair`, `routers/stress_api.py::stress_fail_assign`	System
BR-43	Part Estimation costs a lot's IQC-required parts; "Download & Attach Estimate" raises a whole-lot production sourcing request.	`routers/part_estimation.py` (module docstring)	Production Manager

2.4 QC (Stress Test)

ID	Rule	Source	Role
BR-44	The Stress Test checklist has 27 fixed items; only Touch Screen may be marked N/A.	`models/stress.py::STRESS_CHECKLIST_ITEMS`, `STRESS_OPTIONAL_ITEMS`	QC Inspector
BR-45	Stress Fail must assign an L1 or L2 engineer and sends the tag to L1/L2 with the failed items saved as Stress Notes.	`routers/stress_api.py::stress_fail_assign`	QC Inspector
BR-46	Stress Complete sends the tag to Cosmetic Received and clears old Stress Notes; an assignee is optional.	`routers/stress_api.py::stress_complete_to_paint`	QC Inspector
BR-47	Each Stress result gives the tester a completed WorkID and saves one checklist report (with PDF) per visit.	`routers/stress_api.py::_record_checklist_result`, `stress_fail_assign`	QC Inspector
BR-48	On the legacy QC form, a score of 70/100 or more passes, and the 3rd fail is noted "recommend scrap".	`routers/qc.py::qc_submit`, `QC_FAIL_SCRAP_THRESHOLD`	QC Inspector

2.5 Cosmetic

ID	Rule	Source	Role
BR-49	The cosmetic chain runs in this order: Cosmetic Received, Cleaning, Putty, Dry Sanding, Masking, Painting, Water Sanding, Cosmetic Completed, Final QC.	`routers/cosmetic.py::COSMETIC_PIPELINE`, `NEXT_COSMETIC`	Cosmetic Lead
BR-50	Every cosmetic Move needs an assignee, except a move that lands on Cosmetic Completed or Final QC.	`routers/cosmetic.py::advance_stage`	Cosmetic Lead
BR-51	If the mover appears in a saved Flow Data row, the next stage's user is assigned automatically.	`routers/cosmetic.py::_resolve_flow_next_user`	Cosmetic Lead
BR-52	A tag can skip the rest of the cosmetic chain and go straight to Final QC from any cosmetic stage.	`routers/cosmetic.py::advance_stage`	Cosmetic Lead
BR-53	Cosmetic Fail is allowed only at Cosmetic Received, Cleaning, Water Sanding and Cosmetic Completed, and must assign an L1/L2 engineer.	`routers/cosmetic.py::cosmetic_fail_assign`	Cosmetic Lead
BR-54	Bulk Assign is for admin and Cosmetic Manager only; at Cosmetic Received it also moves tags to Cleaning.	`routers/cosmetic.py::bulk_assign`	Cosmetic Lead
BR-55	Each cosmetic page needs its own edit permission (for example `cosmetic_painting`).	`routers/cosmetic.py::PERM_MODULE_BY_STAGE`	Cosmetic Lead
BR-56	Only admin and Cosmetic Manager can create or edit Flow Data rows.	`models/cosmetic_flow.py::CosmeticFlowRow`, `routers/cosmetic.py::flow_data_allowed`	Cosmetic Lead

2.6 Final QC and grading

ID	Rule	Source	Role
BR-57	Final QC decisions need edit permission on `cosmetic_finalqc`.	`routers/cosmetic.py::advance_stage`, `fqc_move_passed`, `fqc_move_failed`	Final QC Inspector
BR-58	Final QC Pass sets the grade (A, B, C, D, E or Scrap) and holds the tag in "Pass Hold".	`routers/cosmetic.py::advance_stage`, `models/device.py::DeviceGrade`	Final QC Inspector
BR-59	On the first Final QC Pass, the device price becomes buying price + parts consumed + changed parts received; later passes do not add again.	`routers/cosmetic.py::advance_stage`	System
BR-60	Verified RAM and hard-drive changes are written onto the tag's spec at Final QC Pass.	`services/parts_changed.py::apply_spec_changes_on_pass`	System
BR-61	Final QC Fail needs a reason; the tag waits in "Fail Hold" until someone clicks Move.	`routers/cosmetic.py::advance_stage`	Final QC Inspector
BR-62	The fail reason routes the tag: Hardware → L1/L2, Software → Stress Test, Cosmetic → Cosmetic Received, each to the last person who worked that step.	`routers/cosmetic.py::FAIL_REASON_TO_DEPT`, `_resolve_fail_engineer`, `fqc_move_failed`	Final QC Inspector
BR-63	A Hardware fail with no engineer found goes to TRC Production (Tag Number Allocation).	`routers/cosmetic.py::fqc_move_failed`	Final QC Inspector
BR-64	"Move to Inventory" moves Pass Hold tags to Ready to Sale.	`routers/cosmetic.py::fqc_move_passed`	Final QC Inspector → Inventory Manager

ID	Rule	Source	Role
BR-65	"Edit" returns a Pass Hold or Fail Hold tag to Final QC and clears the decision.	`routers/cosmetic.py::fqc_edit_revert`	Final QC Inspector
BR-66	Touch Type is saved on both Pass and Fail; a blank value shows as Non-Touch.	`routers/cosmetic.py::advance_stage`, `models/device.py::touch_type`	Final QC Inspector

2.7 Spare parts

ID	Rule	Source	Role
BR-67	Engineers raise part requests as New, Replace or Downgrade; the request is linked to the tag's open WorkID.	`routers/part_requests.py::create_part_request`	Repair Engineer
BR-68	Bulk Part Request raises one request per L1/L2 tag that needs the part and skips tags that already have a live request for it.	`routers/repair.py::bulk_part_request`	Repair Engineer
BR-69	The handover quantity always equals the requested quantity; it cannot be edited.	`routers/part_requests.py::handover_part`, `bulk_request_action`	Spare Parts Manager
BR-70	Bulk Handover acts only on requests still at "requested".	`routers/part_requests.py::bulk_request_action`	Spare Parts Manager
BR-71	Deleting a part request is a soft close ("cancelled"); handed-over or received requests cannot be cancelled.	`routers/part_requests.py::bulk_request_action`	Spare Parts Manager
BR-72	"Procure" creates a sourcing request that the Sales Manager closes in the CRM.	`routers/part_requests.py::procure_part`, `close_sourcing`	Spare Parts Manager / Sales Manager
BR-73	A sourcing request can be verified only after its deal is closed; verifying adds the sourced quantity to stock and reopens the part request.	`routers/part_requests.py::verify_sourcing`	Spare Parts Manager
BR-74	A faulty part under supplier warranty goes Warranty Sent → Warranty Replaced, and the replacement goes back into stock.	`routers/part_requests.py::warranty_replace`, `warranty_received`	Spare Parts Manager
BR-75	Any signed-in user can mark a part PNA on a tag; clearing keeps the history.	`routers/part_requests.py::mark_part_pna`, `models/pna_part.py`	Repair Engineer
BR-76	Parts GRN (As New) records PO, invoice, e-way bill and vehicle details and adds line quantities to Part Master stock.	`routers/parts_grn.py::grn_create`	Spare Parts Manager
BR-77	A harvest part (from an old device) is recorded as an internal Parts GRN ("Harvest") and added to Part Master with source "harvest".	`routers/parts_grn.py::harvest_part`	Spare Parts Manager

2.8 Sales and dispatch

ID	Rule	Source	Role
BR-78	A seller can only sell tags whose entity matches their own profile entity; a seller with no entity cannot sell.	`routers/sales.py::create_sale` (`_check_sellable`)	Sales Executive
BR-79	A blocked tag cannot be sold, except through its own block's Sell button.	`services/control_engine.py::validate_sale_allowed`, `routers/sales.py::create_sale`	Sales Executive

ID	Rule	Source	Role
BR-80	Since 19 Oct 2026 a tag can be sold from any stage.	`services/control_engine.py::validate_sale_allowed`	Sales Executive
BR-81	A sale below the device's total cost is allowed but shows a below-cost warning.	`services/cost_engine.py::below_cost_warning_for`	Sales Executive
BR-82	Sale numbers come from one sequence (SALE-0001…) and are drawn only for tags that will actually sell.	`routers/sales.py::create_sale`	System
BR-83	The selling company is matched to the tag's entity and snapshotted on the sale, so invoices never change later.	`routers/sales.py::create_sale`, `routers/invoices.py::_sale_company_snapshot`	System
BR-84	A sale's invoice number replaces the tag's purchase invoice number.	`routers/sales.py::create_sale`	Sales Executive
BR-85	Warranty expiry = sale date + the warranty type's days.	`routers/sales.py::create_sale`	Sales Executive
BR-86	Only admin can delete sales or change a sale date; referenced sales are skipped and the tag returns to the stage it was sold from.	`routers/sales.py::bulk_delete_sales`, `change_sale_date`	Admin
BR-87	Block Tags hold Ready to Sale tags for a staff member or customer; the Block ID is `DDMMYYYY-CONTAINER-0001`, counting per date and container.	`routers/block_tags.py::create_block`, `_next_block_id`, `search_tags`	Inventory Manager, Sales Executive
BR-88	Setting a Ready to Sale price needs the "Set Price" pricing permission.	`routers/sales.py::_role_can_set_price`	Sales Executive / Manager
BR-89	The Gate Pass lists every sale with its Delivery Challan; confirming exit marks the sale "dispatched" and skips ones already dispatched.	`routers/gate_pass.py::confirm_exit`, `bulk_confirm_exit`	Sales Executive / Inventory Manager
BR-90	A telecaller's dispatch request is approved by the Sales Manager on Ready to Dispatch.	`routers/dispatch.py::approve_dispatch`	Sales Executive / Sales Manager

2.9 Returns and credit notes

ID	Rule	Source	Role
BR-91	An internal return needs a sale on file, a reason, a condition and a "Received By" person.	`routers/sales.py::process_return`	Sales Executive
BR-92	Only one return is allowed per sale.	`routers/sales.py::process_return`	Sales Executive
BR-93	A returned tag moves to Stock In at once; the return waits for manager approval.	`routers/sales.py::process_return`	Sales Executive
BR-94	Replace Now sells a Ready to Sale tag at ₹0 to the same customer; it cannot be the returned tag.	`routers/sales.py::process_return`	Sales Executive
BR-95	Only admin or Sales Manager can approve or reject a return; approval sends the tag to IQC (restock) or Scrapped.	`routers/sales.py::approve_return`, `reject_return`, `MANAGER_ROLES`	Managers
BR-96	Warranty status at return is computed by the server from the sale, never typed in.	`routers/sales.py::process_return`	System
BR-97	An External Tag return creates the tag if unknown (lot EXT-RETURNS) and a ₹0 placeholder sale, and records the paid-repair amount.	`routers/sales.py::process_external_return`	Sales Executive

ID	Rule	Source	Role
BR-98	Return Stock "Verify" records repair and labour cost; "Complete" re-sells the tag to the original customer under a new sale number.	`routers/stock.py::return_stock_verify` , `return_stock_complete`	Inventory Manager
BR-99	A credit note needs a Debit Note number and a "Received By" person, and marks the tag "Return for Credit Note".	`routers/sales.py::process_credit_note`	Sales Executive
BR-100	Credit notes move forward one step at a time through six stages; "Payment Invoice Done" needs a payment invoice number.	`routers/sales.py::_try_advance_return` , `models/sales.py::CN_STAGES`	Accounts / Sales Manager (`credit_note` edit)

2.10 Accounts

ID	Rule	Source	Role
BR-101	Supplier payments and customer receipts can be added only by admin, Inventory Manager or Sales Manager, and the amount must be above zero.	`routers/accounts.py::create_supplier_payment` , `FINANCE_ROLES`	Accounts
BR-102	The Tax Invoice and Delivery Challan print from the sale's frozen company snapshot.	`routers/invoices.py` (`/print` , `/waybill`)	Accounts
BR-103	Lot P&L and other financial reports need the financial report roles.	`routers/reports.py::_require_financials`	Accounts / Managers
BR-114	Buying price, cost, margin and P&L are shown only to management roles; other roles see selling price only where their work needs it (policy 1 Nov 2026; not yet enforced by default).	Business policy, 1 Nov 2026; Security & Access Design, Pricing and financial data policy (`can_view_pricing` , `_require_financials`)	Admin / Management
BR-104	Daily Stock reports are available as overall, tag, location and daily workbook exports.	`routers/reports.py::daily_stock_report` and export routes	Accounts / TRC Manager

2.11 Warranty

ID	Rule	Source	Role
BR-105	A warranty can be set only on a tag that has a sale; the expiry is counted from the sale date.	`routers/extended_warranty.py::extended_warranty_set`	Inventory Manager / TRC Manager
BR-106	An extension date must be later than the current Warranty Stops date; the added days are kept as a running total and noted on the sale.	`routers/extended_warranty.py::extended_warranty_update`	Inventory Manager / TRC Manager

2.12 Scrap

ID	Rule	Source	Role
BR-107	The Tags Scrapped page lists every tag at Scrap for Sale, whatever its grade.	`routers/scrap.py::scrap_products`	Inventory Manager
BR-108	Verifying a scrapped tag's quantity enables its Sell button.	`routers/scrap.py::verify_scrap`	Inventory Manager
BR-109	A Replacement Scrap records which tag replaced it.	`routers/scrap.py::set_scrap_replacement` , `routers/stock.py::scrap_replace_with_another`	Inventory Manager

2.13 Audits and traceability

ID	Rule	Source	Role
BR-110	A physical audit records the stage and location seen against the system values; a tag can be audited only once per report date.	`routers/physical_audits.py::submit_audit`, `_already_audited`	TRC Manager
BR-111	Every stage move should write a StageMovement row; the previous open row gets an exit time.	`models/device.py::StageMovement`, all stage-move endpoints	System
BR-112	Only admin can edit the allowed-transitions table.	`routers/stage_control.py::add_transition`, `remove_transition`	Admin
BR-113	Each WorkID gets a Completed Date when the tag leaves that person's step.	`services/work_order_close.py`, `routers/repair.py::_close_l1l2_work_orders`	System

3. Standard operating procedures

Short daily routines. Screens are named as they appear in the app.

3.1 IQC Inspector

1. Open the lot (or create it) and register incoming tags, or use IQC → New.
2. Fill the IQC form: tag number, lot, specs, grade and location. Add the GRN number if it is known.
3. Run OxyQC on the device if the agent is used, and check the result reached IQC.
4. If tags were registered at GRN Receipt, click **Advance GRN to IQC**.
5. Before leaving, check the IQC list for tags without a GRN number and hand them to the Inventory Manager for mapping.

3.2 Inventory Manager

1. Map or validate GRNs on **GRN Post-IQC** so IQC tags move to Stock In.
2. Review **Inventory Manager (Stock)**: bucket new tags, set location, and flag urgent tags (all four flag fields).
3. Move tags to production: **Move to Production** (bucket) or **Move to L1/L2** (single or bulk). Only Stock In tags move.
4. Use **Transfers** for floor moves and assignments (Finish Good, As-Is Lot, Scrap for Sale).
5. On **Final QC → Devices Passed**, click **Move to Inventory** so tags reach Ready to Sale.
6. Work the **Return Stock** table: verify repair and labour cost, then Complete or rebucket.
7. Review **Tags Scrapped**: verify quantity and record replacements.
8. Create or release **Block Tags** when asked; set or extend warranties on **Extended Warranty**.

3.3 Production Manager

1. Open **TRC Production** and read the tiles (At You, L1/L2, L3/L4, Stress, Cosmetic, Final QC, Tags Returned).
2. Assign each bucket or tag to a department and engineer (L1/L2 Repair, Stress Test, Cosmetic Repair).
3. Clear the **Tag Number Allocation** queue, including Final QC Hardware fails with no engineer.
4. Use **Change Engineer** for absent staff; the old WorkID closes and a new one opens.
5. Review **PNA summary** and stuck tags; chase parts with the Spare Parts Manager.

6. For new lots, run **Part Estimation** and attach the estimate to raise the parts request.
7. Check **WorkID Status** for team output at the end of the day.

3.4 Repair Engineer (L1/L2 and L3/L4)

1. Open **L1/L2 Repair** (or **L3/L4**). Click **Pick This** on your next tag.
2. Click **Start Repair**.
3. Raise part requests from the tag; mark parts PNA if they cannot be sourced.
4. When a part is handed over, **Verify** it on the device page.
5. If deeper repair is needed, **Request to L3/L4** with a repair note.
6. L3/L4: finish with **Mark Complete** (choose the activity) or **Scrap** (Normal or Replacement).
7. L1/L2: clear all PNA marks, then **Mark Complete** to send the tag to Stress Test. After a scrap decision, use **Back to Inventory**.

3.5 QC Inspector (Stress Test)

1. Open **Stress Test** and take the next tag.
2. Install the OS and run the diagnostic or the stress runner if used.
3. Tick each of the 27 checklist items as Pass, Fail (or N/A for Touch Screen).
4. All pass: click **Complete**. The tag goes to Cosmetic Received.
5. Any fail: click **Fail**, pick an L1/L2 engineer and add a note. The tag goes back to L1/L2.

3.6 Cosmetic Lead

1. Open **Cosmetic & Paint** and check each stage tab.
2. On Cosmetic Received, **Bulk Assign** the day's tags to the Cleaning team (or Move one at a time).
3. Each stage worker clicks **Move** when done and picks the next person (or Flow Data picks them).
4. Use **Skip to Final QC** for tags that need no further cosmetic work.
5. If a tag has a hardware problem at Received, Cleaning, Water Sanding or Completed, click **Fail** and assign an L1/L2 engineer.
6. Keep Flow Data rows up to date when teams change.

3.7 Final QC Inspector

1. Open **Final QC** and take the next tag.
2. Run the manual diagnosis. Correct specs if needed (or use Revalidate IQC).
3. Pass: choose the grade, touch type and bucket name, add pass notes, submit.
4. Fail: choose the reason (Hardware, Software or Cosmetic), add notes, submit.
5. On **Devices Failed**, click **Move** or **Bulk Move** to send each tag back.
6. On **Devices Passed**, click **Move to Inventory** to hand tags to the Inventory Manager.

3.8 Spare Parts Manager

1. Open **Part Master → Requests**. Hand over parts in stock (bulk handover for many).
2. Mark **Not In Stock** or **Procure** for parts you cannot supply.
3. On **Sourcing Requests**, verify closed deals and confirm production requests.
4. Record new stock on **Parts GRN** (As New) with PO, invoice and e-way bill.

5. Record parts taken from old devices with **Add Harvest Part**.
6. Handle faulty parts: Scrap, or send under warranty and receive the replacement.

3.9 Sales Executive

1. Review **Ready to Sale**; check prices and blocked tags.
2. Block tags for a customer or colleague on **Block Tags** when asked.
3. Record sales on **New Sale** / **New Tag Sale**, or **Sell** from a block. Check entity warnings.
4. Print the invoice and Delivery Challan; confirm exit on **Gate Pass**.
5. Log customer returns on **Return New**: Internal Tag, External Tag (paid repair) or Credit Note. Use Replace Now for replacements.
6. Raise dispatch requests (telecallers) and review the sales report.

3.10 Accounts

1. Record **Customer Receipts** against sales and **Supplier Payments** against lots or POs.
2. Upload invoices on POs and quotes; complete payments.
3. Advance **Credit Notes** through Debit Note Verified, Verify Payment and Payment Invoice Done.
4. Check the **Sales** list and the **Daily Stock** report.

3.11 Managers / Admin

1. Read the **Dashboard** and key reports (Lot P&L, Sales, Stage Movement).
2. Approve or reject **Pending Returns**.
3. Handle **Tickets** raised by users.
4. Maintain users, roles, Master Data and the Module Permission matrix.
5. Admin only: correct sales (delete or change date) and edit **Stage Control** transitions.

3.12 TRC Manager

1. Review TRC Production, Inventory and Part Master to see the whole floor.
2. Check the **Daily Stock** report.
3. Run **Physical Audits** for the day's stages and locations; resolve mismatches.
4. Set or extend warranties on **Extended Warranty** when approved.
5. Escalate stuck tags (PNA, long WorkIDs) to the Production and Spare Parts Managers.

4. Open questions

#	Question	Code fact	Business view
OQ1	Should sales be limited to Ready to Sale?	Any stage can be sold, including a tag already Sold (`services/control_engine.py::validate_sale_allowed` , `routers/sales.py::create_sale`)	Sales Executive sells Ready for Sale items
OQ2	What happens to a rejected return?	The tag is already in Stock In and stays there; the `returned` stage is never used (`routers/sales.py::process_return` , `reject_return`)	Return, paid repair, replacement or credit note

#	Question	Code fact	Business view
OQ3	Who are Production Manager, Final QC Inspector, Accounts and TRC Manager in the system?	No built-in roles; Production Manager = `inventory_manager` ; Accounts pages hard-code `admin` , `inventory_manager` , `sales_manager` , so a custom Accounts role is refused (`routers/accounts.py::FINANCE_ROLES`)	12 distinct roles
OQ4	Should IQC Inspectors run GRN?	`/grn` allows only `admin` and `inventory_manager` (`routers/grn.py`)	IQC Inspector owns GRN, Lot, IQC
OQ5	Should stage moves check the current stage and the user's role?	Many moves accept any signed-in user and any current stage (for example `routers/repair.py::l1l2_complete_to_stress` , `routers/stress_api.py::stress_fail_assign` , `routers/stock.py::move_to_stock`)	Only the owning role moves a tag forward
OQ6	How are e-way bills generated for sales?	E-way bill fields exist only on inbound Parts GRN and lot GRN; no sale e-way bill flow found	Accounts generates e-way bills
OQ7	Where are rentals handled?	No rental flow found in sales, dispatch or gate pass	Sales Executive handles rentals
OQ8	Should a scrap sale move tags out of Scrap for Sale?	Selling a Scrap for Sale batch marks the batch sold only (`routers/scrap_for_sale.py::sell_scrap_for_sale`)	Inventory Manager handles scrapped tags
OQ9	Should harvest parts link to the source tag?	Harvest records a free-text lot number, not the device (`routers/parts_grn.py::harvest_part`)	Harvest parts come from old devices
OQ10	Which grade scale is official?	Device grade A–E/Scrap; legacy QC check grade A/B/C/D/S (`models/qc.py`)	Final QC assigns grading
OQ11	Should Block Tags be audited and stage-checked on the server?	No audit on create/release; Ready to Sale is checked only in the search box (`routers/block_tags.py::create_block`)	Block tags for any staff
OQ12	Should part request actions check the current status?	Handover, Not In Stock, Procure and Scrap do not check status on single rows (`routers/part_requests.py`)	Spare Parts Manager actions requests
OQ13	Is dispatch approval required before a telecaller sells?	The rule is in the dispatch module text; `create_sale` does not check it	—